6 **(a)** State the relationship between electric potential at a point and electric field strength at the point.

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.....

.....[2]

(b) Two identical solid metal spheres A and B, each of radius 10 cm, are held in a vacuum such that the separation of their centres is 60 cm, as shown by Figure 6.1.

The charge on sphere A is +6.0 nC and the charge on sphere B is +3.0 nC.

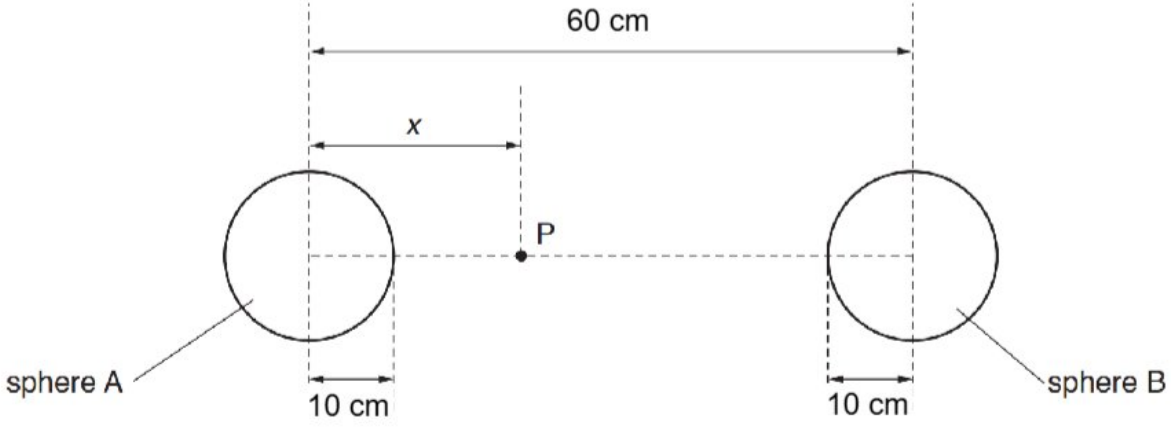


Figure 6.1

Assume that the charge on each sphere is a point charge at the centre of the sphere.

A movable point P is located on the line joining the centres of the two spheres.
The point P is a distance *x* from the centre of sphere A.

(i) On the axes of Figure 6.2, sketch a graph to show the variation with x of the electric potential V between the centres of the two spheres for $x = 0$ to $x = 60$ cm. Use the space above Figure 6.2 for any working.

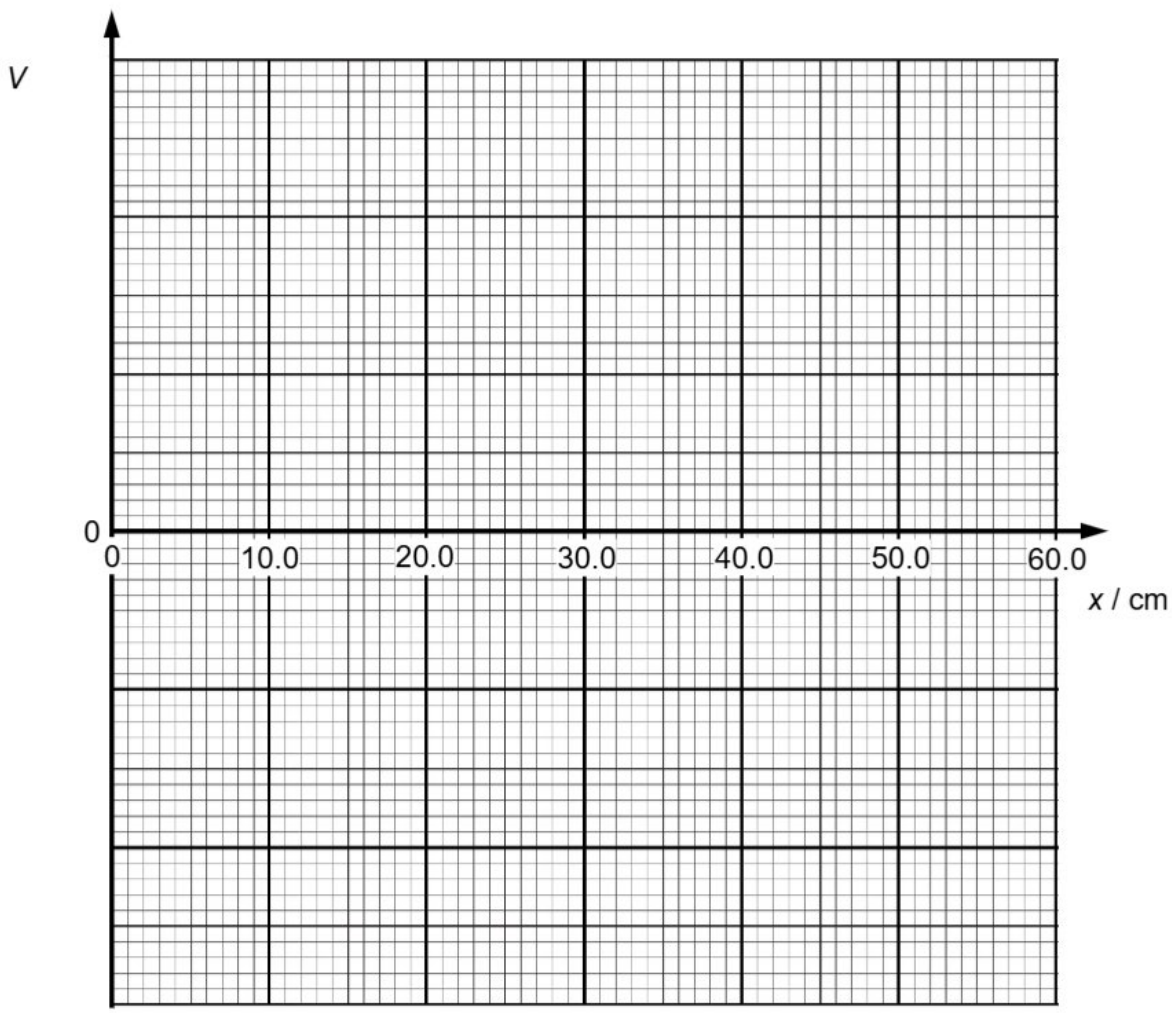


Figure 6.2

[4]

(ii) Calculate the electric potential energy E_P of the two spheres.

E_P = ... J [2]

(iii) The two spheres are now released simultaneously so that they are free to move.

Describe and explain the subsequent motion of the spheres.

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..... [2]

[Total: 10]